Creative Question (Answer any two)

1. Given,

$A=\{x\in N: x^2-5x+6=0\}$
$B=\{x\in N: x \text{ is a prime number and } x<11\}$
$C=\{3,4,5,7,9\}$

(a) Express the set A in roster form.
(b) Show that the number of elements of $P(B\cap C)$ supports 2^n, where n is the number of elements of $B\cap C$.
(c) If $f(x)=\frac{3x+1}{3x-1}$, find $\frac{f\left(\frac{1}{x^2}\right)+1}{f\left(\frac{1}{x^2}\right)-2}$.

2. Given,

$R=\{(x,y): x\in A, y\in A, x-y+2=0\}$
where $A=\{-3,-2,-1,0,1\}$.

(a) Express the set A in set-builder form.
(b) Express the relation R in roster form and find its domain.
(c) Show that the given function satisfies the required relation.

3. Given,

$A=\{x: x\in N, x^2>7, x^3<190\}$,
$B=\{x: x\in Z, x^2\le 4\}$ and
$S=\{(x,y): x\in B, y\in B, y-x=-1\}$.

(a) Express $C=\{y\in N: y^2+3y+2=0\}$ in roster form.
(b) Find P(A).
(c) Express the relation S in roster form and find its domain

MCQ (Answer all the question)

1. If $A=\{2,3,4\}$ and $B=\{1,3\}$ which of the following is correct?

i. $A\cap B=\{3\}$
ii. $A-B=\{2,4\}$
iii. $A\times B$ has 5 elements.

A. i and ii **B.** i and iii
C. ii and iii **D.** i, ii and iii

2. Let $A=\{3,5,7\}$ and $B=\{4,5,7\}$ Which of the following is correct?
i. $A\cap B=\{5,7\}$
ii. $P(A\cup B)$ has 16 elements.
iii. $A-B=\{3,4\}$

A. i and ii **B.** i and iii
C. ii and iii **D.** i, ii and iii

3. Which of the following represents the set $\{x\in N: x \text{ is a prime number and } 2\le x\le 7\}$ in roster form?
A. {2,3,7} **B.** {2,3,7}
C. {3,5,7} **D.** {2,3,5,7}

4. If $X=\{a,b,c\}, Y=\{b\}, Z=X-Y$ then how many elements does P(Z) have?
A. 1 **B.** 2
C. 3 **D.** 4

5. If $Q=\{1,a\}$, which of the following is P(Q)?
A. {1,a} **B.** {{1},{a},{1,a}}
C. {{1},{a},{1,a},∅} **D.** {{1,a}}

6. Let $D=\{y: y\in N, 5\le y\le 10\}$. Which is the roster form of D?
A. {5,6,7,8,9,10} **B.** {6,7,8,9}
C. {5,6,7,8,9} **D.** {6,7,8,9,10}

7. If $A=\{2,3,4\}, B=\{3,4,5\}$, then A−B is—
A. {2} **B.** {5}
C. {3,4} **D.** {2,3,4,5}

8. If $F(y)=y^4+5y^2-3$, then what is F(−1)?
A. −9 **B.** −7
C. 3 **D.** 4

9. Let $A=\{3,5,7\}$ and $B=\{4,5,7\}$. Which statement is correct?
i. $A\cap B=\{5,7\}$
ii. $n[P(A\cup B)]=16$
iii. $A-B=\{3,4\}$
A. i and ii **B.** i and iii
C. ii and iii **D.** i, ii and iii

10. If $P=\{2,4,6,8\}$, then how many proper subsets does P have?
A. 4 **B.** 8
C. 15 **D.** 16

11. If $M=\{2,3,4\}$ then how many elements does P(M) have?
A. 3 **B.** 6
C. 7 **D.** 8

12. If $A=\{a,b,c\}$ how many proper subsets does A have?
A. 3 **B.** 7
C. 8 **D.** 9

13. Which of the following relations is a function?
A. {(2,3),(2,5)} **B.** {(−1,2),(−1,4)}
C. {(2,3),(1,2)} **D.** {(−4,3),(−4,4)}

14. If $M=\{1,2\}, N=\{-1,0,1\}$ then—
i. The number of proper subsets of M is 3.
ii. $M\cap N=\{1\}$
iii. $N-M=M$
Which is correct?
A. i and ii **B.** i and iii
C. ii and iii **D.** i, ii and iii

15. Consider the Venn diagram where the elements of P are 1,2,4,5 and the elements of Q are 2,3,5,6. Then P−Q= ?
A. {2,5} **B.** {1,4}
C. {3,6} **D.** {1,2,4,5}

Short Question (Answer all the Question)

1. If $U=\{1,3,5,7,9,11,13\}, A=\{9,11,13\}$ find A'.

2. If $B=\{5,7,11,13\}$ find P(B) and show that the number of elements of P(B) is 2^n, where n is the number of elements of B.

3. Express the following set in roster form:
$A=\{x\in N: x \text{ is a prime number and } 2\le x\le 7\}$.

4. If $P=\{1,3,4\}, Q=\{2,5,7,9\}$ what type of sets are P and Q with respect to each other

5. Write the two methods of representing a set.

LESSON BY SHAFIQ